



Industry-Academia Consortium in Control Energy and Measurement

13th May 2024 to 17th May 2024

“In association with HRDC, IEM UEM Group”

Jointly Organized by Department of Electrical Engineering and Electrical & Electronics Engineering,
Institute of Engineering & Management, Kolkata

Technically sponsored by IET, IEI

The objectives of this FDP are

The FDP aims to foster collaboration between academia and industry in the areas of control systems, energy management, and measurement skills. Through this program, participants will gain valuable insights into the latest industry trends, practical applications, and innovative techniques in these crucial domains.

Organizing Committee Member:

1. Prof. Pratik De Sarkar (Assistant Professor, EEE dept, IEM, Kolkata)
2. Prof. Unmesha Ray (Assistant Professor, EE dept, IEM, Kolkata)
3. Prof. Bed Prakash Das (Assistant Professor, EE dept, IEM, Kolkata)

Participants:

Nine (9) eminent speakers from various renowned institutes across the Globe and around **150** participants from different institutes in India.

Day wise Schedule:

Day 1: 1st Half

Date: 13th May 2024 **Session 1: Time:** 10:00AM – 12:00 PM

Speaker: Dr. Runu Banerjee

Professor, Jadavpur University

Title of the talk: Electronic Nose- A Smart Sensing Technique for Food Quality Estimation based on Aroma Detection

Day 1: 2nd Half

Date: 13th May 2024 **Session 2: Time:** 2:00PM – 4:00 PM

Speaker: Dr. Karamjyoti Panigrahi

Senior Process Engineer, OLA (Electric) Cell Technology Private Limited

Title of the talk: Fueling the Future: Insights into Materials, Process, and Measurements Driving EV battery advancements

Day 2: 1st Half

Date: 14th May 2024

Session 3: Time: 10:00AM – 12:00 PM

Speaker: Mr. Poulastya Mukherjee

Robotics Engineer, Energy Robotics, Germany

Title of the Talk: AI Enabled Business Analytics for Industry 4.0

Day 2: 2nd Half

Date: 14th May 2024

Session 4: Time: 2:00 PM– 4:00 PM

Speaker: Dr. Srijan Sengupta

Associate Professor, IIT, Jodhpur

Title of the talk: Batteries beyond Lithium chemistry

Day 3: 1st Half

Date: 15th May 2024

Session 5: Time: 10:00AM – 12:00 PM

Speaker: Dr. Aritro Dey

Assistant Professor, NIT, Durgapur

Title of the Talk: Flight Control Systems: An Overview

Day 3: 2nd Half

Date: 15th May 2024

Session 6: Time: 2:00 PM– 4:00 PM

Speaker: Dr. Binoy Kr Karmakar

Assistant Professor, Calcutta University

Title of the talk: Smart Home Energy Management

Day 4: 1st Half

Date: 16th May 2024

Session 7: Time: 10:00 AM– 12:00 PM

Speaker: Dr. Saikat Ch. Bakshi

Assistant Professor, NIT, Calicut

Title of the Talk: National Education Policy 2020

Day 5: 1st Half

Date: 17th May 2024

Session 8: Time: 10:00 AM– 12:00 PM

Speaker: Dr. Bhaswar Chakrabarti

Assistant Professor, IIT, Madras

Title of the talk: Volatile resistance switching in memristive devices for dendritic computation

Day 5: 2nd Half

Date: 17th May 2024

Session 9: Time: 3:00 PM– 4:00 PM

Speaker: Mr. Basumitra Dutta

Manager(Elect), IPH, Damodar Valley Corporation (DVC)

Title of the talk: Electrical Maintenance aspects of Generator and Auxiliaries in a Thermal Power Plant

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The negative electrode half-reaction for the graphite:

$$\text{LiC}_6 \rightleftharpoons \text{C}_6 + \text{Li}^+ + \text{e}^-$$

The positive electrode half-reaction is the attachment:

$$\text{CoO}_2 + \text{Li}^+ + \text{e}^- \rightleftharpoons \text{LiCoO}_2$$

The full reaction being:

$$\text{LiC}_6 + \text{CoO}_2 \rightleftharpoons \text{C}_6 + \text{LiCoO}_2$$

3 V wet batteries
1.2 V wet batteries
Voltage = 1.2 - (-3) = 4.2 V

Total weight
= weight of LiCoO_2 + weight of C
= 91 + 6 x 12 = 163 gm

Total charge = 96500 coulomb
= 96500/3.6 mAh
= 26805.56 mAh

Specific cap. = 26805.56/163
= 164.45 mAh/gm

Specific energy = 3.7*164.45
= 608.47 Wh/Kg

Energy_{wt} = 225 Wh/Kg

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